

https://github.com/aarhusuni/Week6_Assignment/tree/main/tests

In the assignment I decided to test Den Gammleby's website: <https://www.dengamleby.dk/>

1. Manual testing

I inspected Den Gammleby website for accessibility violations. Navigated the website with NVDA and keyboard.

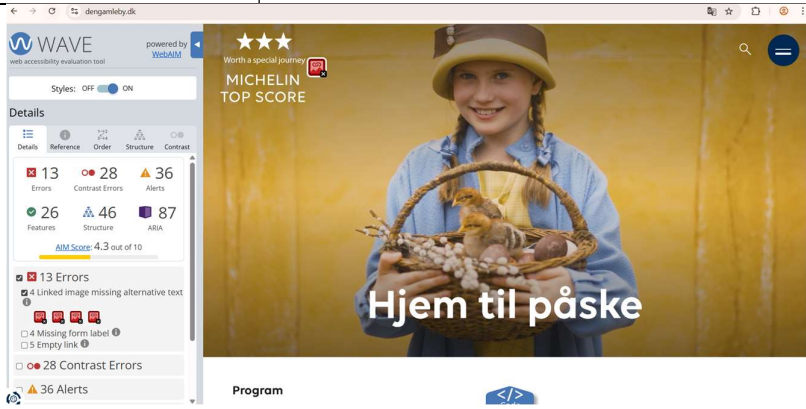
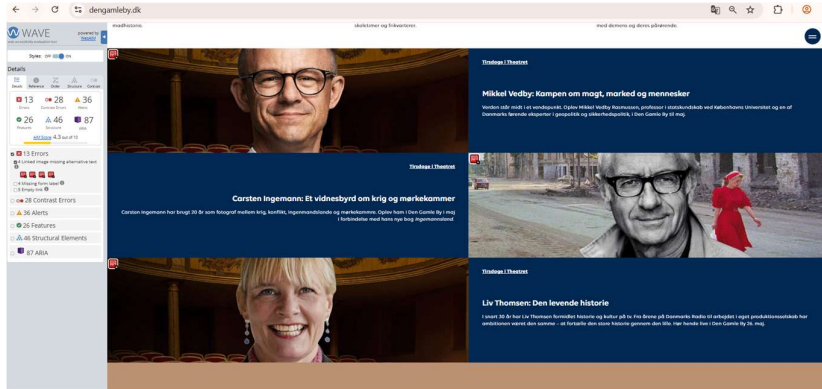
- language and search and the menu is not focusable with screen reader
- the slider is a cool design feature but when it announced its quite confusing
- under the den Gammelby header the three-star scoring is also a link leading to the homepage, unnecessary and confusing with the reader
- pictures are navigable and their announced titles are not too corresponding to the picture
- “Home for Easter” it is announced it's the main landmark, and that it's a link but the name (Home for easter) or any information where it navigates is not announced
- if we click the Home for Easter link leads to a page about gammlebys easter programs, nothing is navigable on the page.

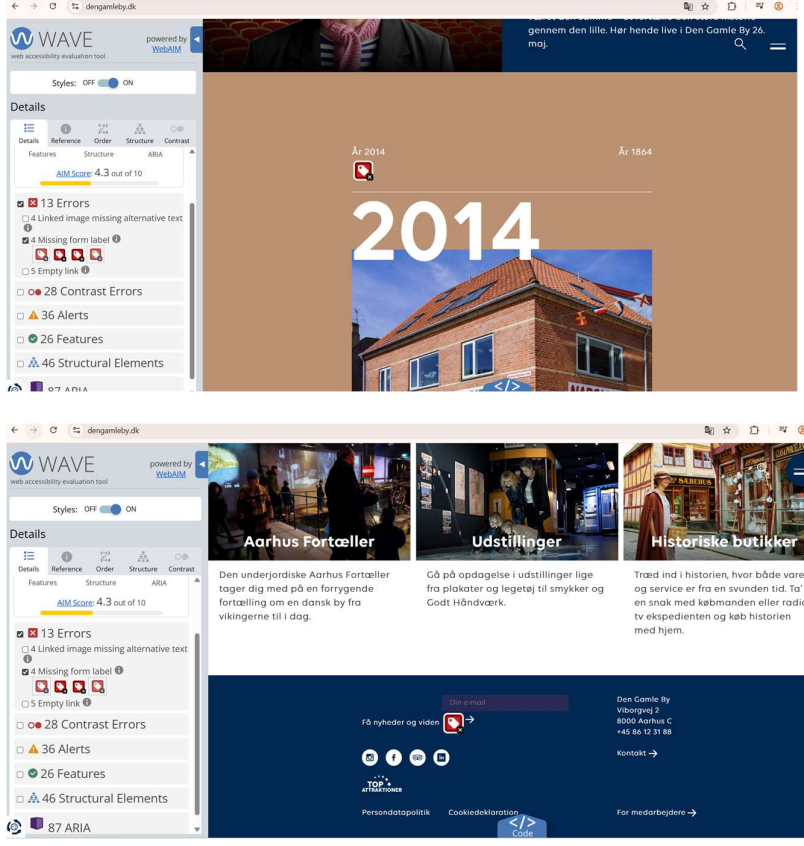
2. Testing with Wave and axe DevTools, Bug report

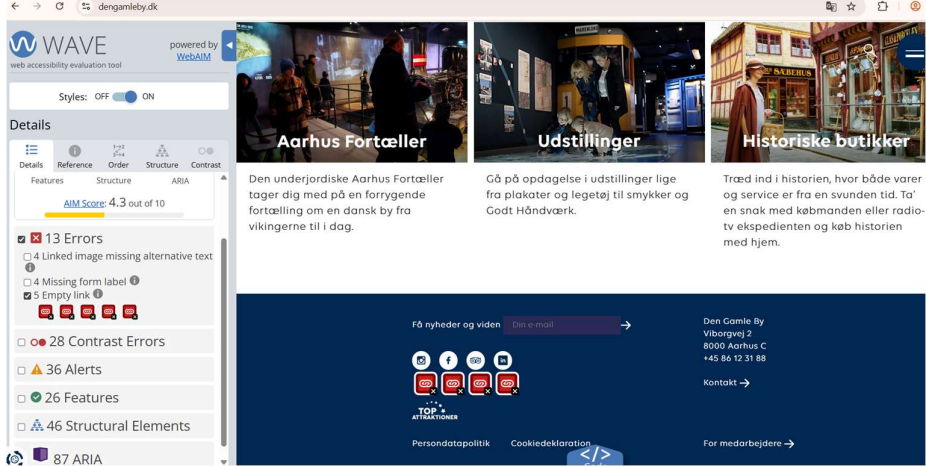
Issues found:

WAVE: 12 error, 28 contrast error

Axe DevTool: 19 issues

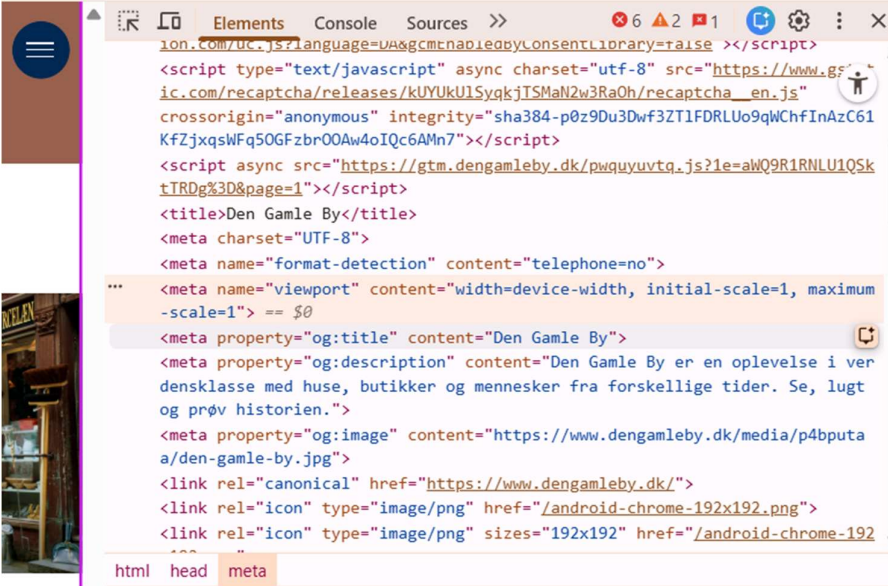
BugID:	001
Title:	Images miss alternative text
Description	Image is in a link but describing text of the image is not present, hence screen reader doesn't reads it so the user doesn't know where it leads.
Steps to reproduce:	- go to website: https://www.dengamleby.dk/ - navigate with screen reader to different images Bug was found using WAVE.
Expected result:	Descriptive titles of images are announced, so if the picture works as a link (the picture is navigable) then the user know where it leads.
Actual result:	No image name announced that describes where the images leads.
Severity:	major
Priority:	heigh
Environment:	Google Chrome Version 146.0.7680.165 (Official Build) (64-bit)
Attachments:	 
Reported by:	Csaba Pákozdi
Date:	02/04/2026

BugID:	002
Title:	Missing label
Description	Form control (elements that let the user enter data) doesn't have label.
Steps to reproduce:	1. go to website: https://www.dengambleby.dk/ 2. navigate with screen reader to slider, or the bottom of the page the subscription to newsletter 3. screen reader doesn't announce these form controls
Expected result:	Screen reader announces the purpose of form controls.
Actual result:	No information about the purpose of these form controls, for example its announce there is a slider there, but for what remains a mystery..
Severity:	major
Priority:	heigh
Environment:	Google Chrome Version 146.0.7680.165 (Official Build) (64-bit)
Attachments:	
Reported by:	Csaba Pákozdi
Date:	02/04/2026

BugID:	003
Title:	A link contains no text.
Description	Links that are not announced correctly
Steps to reproduce:	1. go to website: https://www.dengamleby.dk/ 2. navigate "spisested" in the menu or the links leading to social medias
Expected result:	Screen reader announces where certain links lead
Actual result:	No information about where link leads.
Severity:	major
Priority:	heigh
Environment:	Google Chrome Version 146.0.7680.165 (Official Build) (64-bit)
Attachments:	
Reported by:	Csaba Pákozdi
Date:	02/04/2026

```
git add .
git commit -m "accessibility"
git push
```

BugID:	004
Title:	Contrast error.
Description	Very low contrast between text and background colors.
Steps to reproduce:	1. go to website: https://www.dengamleby.dk/ 2. there are multiple contrast error, especially text in front of pictures
Expected result:	Heigh contrasted texts so people with visual impairment can also read it
Actual result:	Low contrast especially text in front of pictures and against the brownish background according to WAVE. DevWebtool doesn't show error against the brownish background.
Severity:	major
Priority:	heigh
Environment:	Google Chrome Version 146.0.7680.165 (Official Build) (64-bit)
Attachments:	 Vores tid 2000-2014 Dronning Margrethes Gade er et snapshot af året 2014. Vores tid på museum. 2014-gaden er en sivegade med gadeforretninger og butikcenter, som hedder HoLM1. Her er solcenter, 7-Eleven, bank, Blockbuster, pizzeria og en gammel jazzbar, der lugter af røg og ølsjatter. Kig ind til singlekvinden Rikke og besøg regnbuefamilien Isager. Besøg deres hjem og se interviews, der fortæller om deres helt almindelige liv med børn og arbejde. Tidsrejse i danmarkshistorien →
Reported by:	Csaba Pákozdi
Date:	02/04/2026

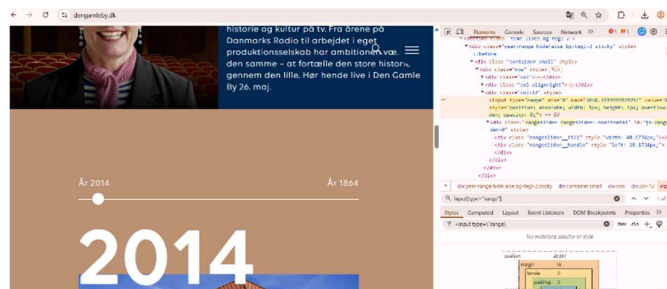
BugID:	005
Title:	Zooming and scaling must not be disabled.
Description	Users with partial vision and low vision often choose to enlarge the fonts on their browser to make text on the web easier to read.
Steps to reproduce:	1. go to website: https://www.dengambleby.dk/ 2. ??
Expected result:	Font enlarges if the user zooms.
Actual result:	Font enlarges if the user zooms? I used DevTool to find this bug, both DevTool and the automated test in Playwright found it, although when I actually zoom on the page with Ctrl+zoom the font enlarges. The description of the bug says the the maximum-scale parameter of the zoom must be 2 or more. Right now its 1, so that could be the case this bug shows, so we can zoom, but not as much as its expected according to WCAG.
Severity:	moderate
Priority:	low
Environment:	Google Chrome Version 146.0.7680.165 (Official Build) (64-bit)
Attachments:	Ensures that the user-scalable="no" parameter is not present in the <meta name="viewport"> element and the maximum-scale parameter is not less than 2. 
Reported by:	Csaba Pákozdi
Date:	02/04/2026

3. Automated testing, Bugreport

The automated test found 11 issues.

3.2 Range doesn't have label

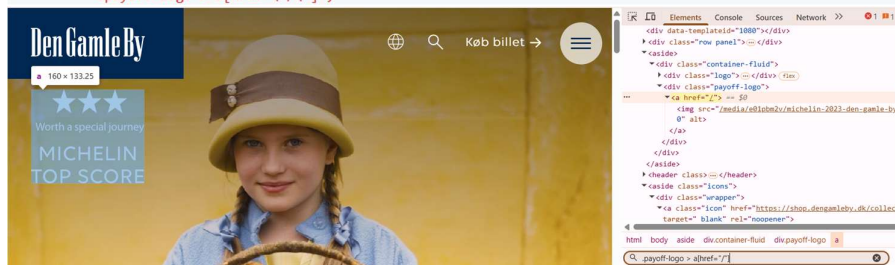
```
+ Array [
+   Object {
+     "description": "Ensure every form element has a label",
+   },
+   ],
+   "failureSummary": "Fix any of the following:
+   Element does not have an implicit (wrapped) <label>
+   Element does not have an explicit <label>
+   aria-label attribute does not exist or is empty
+   aria-labelledby attribute does not exist, references elements that do not exist or references elements that are empty
+   Element has no title attribute
+   Element has no placeholder attribute
+   Element's default semantics were not overridden with role=\"none\" or role=\"presentation\"",
+   "html": "<input type=\"range\" min=\"0\" max=\"3801.1499999999996\" value=\"0\" style=\"position: absolute; width: 100%; height: 100%; border: none; border-bottom: 1px solid black; margin: 0;\" />",
+   "impact": "critical",
+   "none": Array [],
+   "target": Array [
+     "input[type=\"range\"]",
+   ],
+ ]
```



3.3 Links doesn't have describing texts

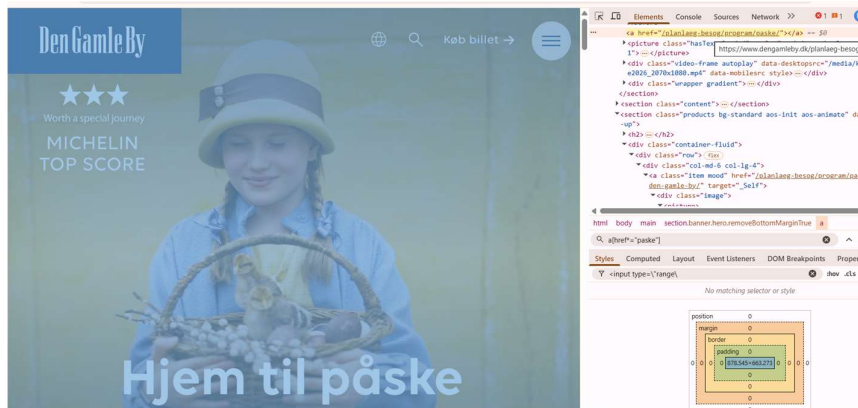
3.2.1 Image1

```
Object {
  "description": "Ensure links have discernible text",
  "help": "Links must have discernible text",
  "helpUrl": "https://dequeuniversity.com/rules/axe/4.11/link-name?application=playwright",
  "id": "link-name",
}
Fix any of the following:
Element does not have text that is visible to screen readers
aria-label attribute does not exist or is empty
aria-labelledby attribute does not exist, references elements that do not exist or references elements that are empty
Element has no title attribute,
"html": "<a href=\"/\"><img src=\"/media/e01pbm2v/michelin-2023-den-gamle-by.svg\" width=\"130\" alt=\"\"></a>",
"impact": "serious",
"none": Array [
  Object {
    "data": null,
    "id": "focusable-no-name",
    "impact": "serious",
    "message": "Element is in tab order and does not have accessible text",
    "relatedNodes": Array [],
  },
],
"target": Array [
  ".payoff-logo > a[href=\"/\"/]",
]
```



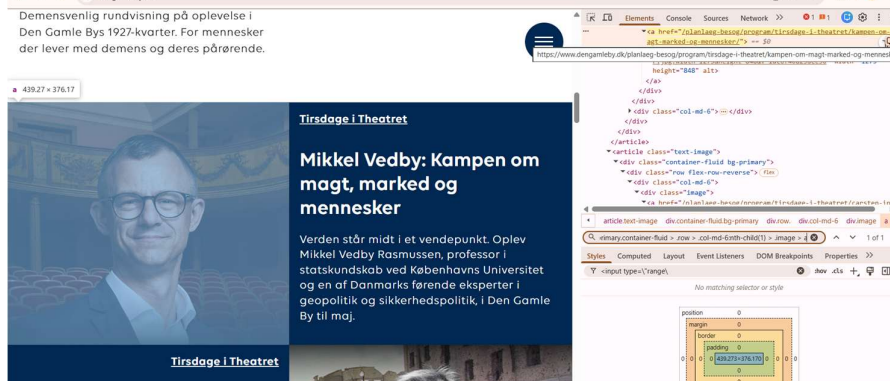
3.2.2 Image2

```
Fix any of the following:
Element does not have text that is visible to screen readers
aria-label attribute does not exist or is empty
aria-labelledby attribute does not exist, references elements that do not exist or references elements that do not have text
Element has no title attribute",
"html": "<a href=\"/planlaeg-besog/program/paske/\"></a>",
"impact": "serious",
"none": Array [
  Object {
    "data": null,
    "id": "focusable-no-name",
    "impact": "serious",
    "message": "Element is in tab order and does not have accessible text",
    "relatedNodes": Array [],
  },
],
"target": Array [
  a[href$=\"/paske/\"],
]
```



3.2.3 Image3

```
+ Fix any of the following:
+ Element does not have text that is visible to screen readers
+ aria-label attribute does not exist or is empty
+ aria-labelledby attribute does not exist, references elements that do not exist or references elements that are empty
+ Element has no title attribute",
+ "html": "<a href=\"/planlaeg-besog/program/tirsdage-i-theatret/kampen-om-magt-marked-og-mennesker/\"></a>",
+ "impact": "serious",
+ "none": Array [
+   Object {
+     "data": null,
+     "id": "focusable-no-name",
+     "impact": "serious",
+     "message": "Element is in tab order and does not have accessible text",
+     "relatedNodes": Array [],
+   },
+ ],
+ "target": Array [
+   .text-image:nth-child(4) > .bg-primary.container-fluid > .row > .col-md-6:nth-child(1) > .image > a",
+ ]
```



There are **2 more** similar image without describing text.

3.4 Element does not have text that is visible to screen readers

The social media links (4 in total) are missing describing elements for screen reader.

```
Fix any of the following:
Element does not have text that is visible to screen readers
aria-label attribute does not exist or is empty
aria-labelledby attribute does not exist, references elements that do not exist or references elements that are empty
Element has no title attribute
  "html": "<a class='instagram' href='https://www.instagram.com/den_gamle_by/' target='_blank' rel='noopener'>
  "impact": "serious",
  "none": Array [
    Object {
      "data": null,
      "id": "focusable-no-name",
      "impact": "serious",
      "message": "Element is in tab order and does not have accessible text",
      "relatedNodes": Array [],
    },
  ],
  "target": Array [
    ".instagram",
  ],
```

3.5 Zooming and scaling must not be disabled

```
Object {
  "description": "Ensure <meta name='viewport'> does not disable text scaling and zooming",
  "help": "Zooming and scaling must not be disabled",
  "helpUrl": "https://dequeuniversity.com/rules/axe/4.11/meta-viewport?application=playwright",
  "id": "meta-viewport",
  "impact": "moderate",
  "nodes": Array [
    Object {
      "all": Array [],
      "any": Array [
        Object {
          "data": "maximum-scale",
          "id": "meta-viewport",
          "impact": "moderate",
          "message": "maximum-scale on <meta> tag disables zooming on mobile devices",
          "relatedNodes": Array [],
        },
      ],
    },
  ],
  "failureSummary": "Fix any of the following:
maximum-scale on <meta> tag disables zooming on mobile devices",
  "html": "<meta name='viewport' content='width=device-width, initial-scale=1, maximum-scale=1'>",
  "impact": "moderate",
  "none": Array [],
  "target": Array [
    "meta[name='viewport']",
  ],
}
```

4. Conclusion

If we compare accessibility bug finding using different methods, we can see there is a significant difference between the number of bugs found. WAVE was better at finding form control bugs, it could also find it in the footer, while axe DevTool and the automated test couldn't. There is also a difference how the tools read a certain bug, for example if a image is a link as well then Axe DevTool puts it in the link missing text category while WAVE and the Automated test put it in the image missing text category. WAVE is much more sensitive to colour contrast bugs, although it couldn't find the zoom disabled. As for the zoom disabled, as I wrote previously, that bug was hard to understand, probably it comes from the fact that WCAG demands *maximum-scale* parameter to be equal or more the 2 but right now its 1. For thorough bug investigation I believe the best practice is to use all the 3 tools.

	Image missing text	Form control	Link missing text	Colour contrast	Zoom disabled	SUM
Wave	4	3	5	28	-	40
Axe Devtool	-	1	9	8	1	19
Automated	5	1	4	-	1	11